

Aanidhya Patidar

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SUMMARY

Computer Science (AI & ML) student focused on scalable backend systems and applied machine learning. Recently built and deployed an agricultural fintech PWA with custom AI localization, real-time WebSocket chat, and sub-zooms API pipelines serving Indian farmers. Seeking a backend or ML internship to solve complex engineering problems.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (AI & ML)

GPA: 8.71/10.0

Bhopal, MP

2023 – 2027

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), SQL

Frameworks & Libraries: FastAPI, Flask, React, Vite, Pandas, NumPy, Scikit-learn, LightGBM, SQLAlchemy, TanStack Query, i18next, Recharts, Sentence-Transformers

Databases: PostgreSQL, SQLite, MySQL

Developer Tools: Git, GitHub, VS Code, Postman, Alembic, Docker, Vercel, Render

PROJECTS

Krishi Khata — Digital Farm Ledger & Market Intelligence PWA | *FastAPI, React, Vite, PostgreSQL, Gemini AI*

2025 – 2026

- Built and deployed a mobile-first PWA for Indian farmers — digital accounting ledgers, real-time mandi price tracking, and AI crop advisory. Frontend on Vercel, backend on Render, backed by Neon PostgreSQL.
- Designed a JIT mandi price pipeline using government APIs, caching 30-day commodity histories per district in PostgreSQL for sub-zooms lookups — replacing wasteful batch scraping entirely.
- Integrated a bilingual AI engine via Google Gemini — the backend intercepts `Accept-Language` headers and rewrites Gemini system instructions at runtime, streaming native Hindi or English responses without storing translated text.
- Set up a hardened WebSocket gateway for real-time community chat with JWT-authenticated handshakes, pre-upgrade rejection of unauthorized connections, and SlowAPI rate limiting on write-heavy routes.
- Localized the React frontend across 70+ UI strings via i18next, added searchable Combobox dropdowns for mandi queries, and built a Quick-Select Crop Hub that auto-fills search parameters from the farmer's active crops.

QueryLens — Natural Language SQL Debugger | *Python, Flask, React, Vite, scikit-learn, Sentence-Transformers, SQLite*

2025

- Built a semantic SQL debugging tool that takes a broken query and database schema as input, identifies *why* the query is semantically wrong (not just syntactically), and generates an educational explanation alongside the corrected query.
- Architected a four-layer pipeline: a Schema Parser that reflects database structure into a graph via SQLAlchemy, a Query Intent Extractor that parses SQL into structured ASTs using `sqlglot`, a Semantic Error Classifier, and an LLM-powered Explanation Generator.
- Developed a typed error taxonomy covering six semantic error classes — wrong join direction, aggregation type mismatch, missing GROUP BY, ambiguous column references, cartesian product risk, and implicit type coercion — enabling precise, structured error signals rather than generic LLM responses.
- Trained a lightweight classifier on 5,000+ synthetically generated (schema, broken query, error_type) triples to detect error categories, feeding structured signals into the LLM for targeted explanations.

CERTIFICATIONS

- Applied Machine Learning in Python** — University of Michigan (Coursera)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**

LANGUAGES

English (Fluent), Hindi (Native)